

Practice Quiz: Learning (Implementation)

Name: __ Date: __

Part A: Multiple Choice

1. Dirichlet concentration parameters encode: A) The exact probability values B) Pseudo-counts that represent accumulated experience — more counts = more confidence C) The number of states D) The temperature of the system
2. The Dirichlet mean $E[\theta_i] = \alpha_i / \sum \alpha$ gives: A) The mode of the distribution B) The expected probability for each outcome, computed from concentration parameters C) The variance D) A random sample
3. The `update_A` method increments concentration parameters by: A) A fixed learning rate B) The outer product of the observation vector and state beliefs C) One everywhere D) A random amount
4. Learning rate decreases because: A) We manually reduce it B) Each new observation contributes $1/N$ to the total, where N grows with experience C) The agent gets tired D) Python slows down
5. A strong Dirichlet prior (large α): A) Learns from every observation equally B) Changes slowly because each new observation is a small fraction of total experience C) Is always better D) Cannot be overridden
6. To implement "forgetting," you would: A) Delete the agent B) Reset concentration parameters to smaller values, reducing accumulated experience C) Increase gamma D) Clear the observation log
7. `update_B` needs both previous and current beliefs because: A) B has 3 dimensions B) It's learning $P(s_t | s_{t-1}, a)$ — the transition from previous to current state C) It updates faster that way D) NumPy requires two arrays

Part B: Short Answer

1. Start with $A_{\text{prior}} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$. After observing state 0 produce observation 0 five times and observation 1 twice, compute the updated A_{prior} and the new A matrix (Dirichlet mean).
2. Explain why a "weak prior" agent learns faster but is initially less accurate, while a "strong prior" agent learns slower but may be initially more accurate. When would you prefer each?
3. Design a test to verify your learning implementation: what properties should the learned A matrix have after sufficient observations from a known environment?